BullSO 2026 Invitational

Water Quality C

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Name:_____

Participant Names:_____

Total Score:_____/100 Rank:_____

Part I: Freshwater Ecology (30 points) (1.5 points each)

1. Which process is primarily responsible for returning nitrogen from the biosphere to the atmosphere?

A. Nitrification

B. Ammonification

C. Denitrification

D. Nitrogen fixation

2. Increased phosphorus input into a freshwater lake most directly leads to:

A. Acidification

B. Eutrophication

C. Thermal pollution

D. Salinization

3. Which of the following best describes a watershed?

A. An underground aquifer that supplies drinking water

B. A region where all precipitation drains to a common outlet

C. A man-made water treatment facility

D. A wetland dominated by emergent vegetation

4. Short Answer: Explain two ways sedimentation can negatively impact freshwater ecosystems.

5. Traditional Ecological Knowledge (TEK) is best described as:

A. Information obtained only through modern laboratory experiments

B. Knowledge developed by Indigenous peoples through long-term interaction with ecosystems

C. Data collected exclusively by government agencies

D. Beliefs unrelated to scientific observation

6. Which trophic level is most directly affected by a sudden drop in dissolved oxygen?

A. Primary producers

B. Primary consumers

C. Secondary consumers

D. Decomposers

7. Fill in the blank: The process of converting wastewater into effluent safe for discharge is known as _____ treatment.

8. Which factor most strongly influences the solubility of oxygen in water?

A. pH

B. Temperature

C. Salinity

D. Turbidity

9. Short Answer: Describe one difference between potable water treatment and wastewater treatment.

10. A population that shows rapid growth, early reproduction, and little parental care is best described as:

A. K-selected

B. Density-independent

C. R-selected

D. Stable

11. In potable water treatment, what is the primary purpose of adding alum (aluminum sulfate)?

A. To cause small particles to clump together

B. To adjust the pH of the water

C. To kill Bacteria and Viruses

D. To add fluoride for dental health

12. The "Schmutzdecke" is a beneficial biological layer found in which type of filter?

A. Slow sand filter

B. Rapid sand filter

C. Reverse osmosis membrane

D. Carbon filter

13 . In the activated sludge process, what is the role of the aeration tank?

A. To settle out heavy solids by gravity

B. To disinfect water using UV light

C. To filter out microplastics

D. To provide oxygen to bacteria so they can consume organic matter

14. Explain the difference between "Coagulation" and "Flocculation."

15. Approximately what percentage of energy is typically transferred from one trophic level to the next?

A. 100%

B. 50%

C. 10%

D. 1%

16. Which process explains why top predators like osprey or eagles often have high concentrations of toxins like DDT or mercury in their bodies?

A. Eutrophication

B. Biomagnification

C. Bioaccumulation

D. Decomposition

17. In most aquatic ecosystems, which group of organisms forms the base of the food web?

A. Zooplankton

B. Small Fish

C. Phytoplankton

D. Benthic invertebrates

18. Which human activity is typically the largest contributor to sediment pollution by volume?

A. Nuclear power generation

B. Construction and agriculture

C. Wastewater treatment effluent

D. Acid mine drainage

19. "Bed load" refers to:

A. Sediment that is dissolved in the water

B. Sediment that floats on the surface

C. Sediment that remains suspended on the water column

D. Sediment that rolls or slides along the bottom of the stream

20. What is the primary negative effect of "impervious surfaces" (like concrete and asphalt) on a watershed?

A. They increase the velocity of stormwater runoff, leading to erosion

B. They reduce the speed of surface runoff, delaying the water cycle

C. They increase the amount of water infiltrating into groundwater, overflowing the water table

D. They prevent filtration of pollutants before they reach the stream

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Part II: Freshwater Macroinvertebrates (30 points)

Common names only.

21. Identify the macroinvertebrate most associated with highly oxygenated, clean water:

A. Tubifex worm

B. Stonefly

C. Blood midge

D. Leech

22. Which class of macroinvertebrates is considered pollution tolerant?

A. Class 1

B. Class 2

C. Class 3

D. Class 4

23. Match the organism to its pollution tolerance class: *(Choices: Class 1, Class 2, Class 3, Class 4, Class 5)*

- A. Mayfly: _____
- B. Dragonfly: _____
- C. Mosquito larva: _____
- D. Water penny: _____

24. Which macroinvertebrate breathes atmospheric oxygen and surfaces periodically?

A. Water strider

B. Riffle beetle larva

C. Caddisfly larva

D. Water mite

25. Short Answer: Why are macroinvertebrates useful indicators of long-term water quality?

26. Which invasive aquatic plant forms dense mats and outcompetes native vegetation?

A. Purple loosestrife

B. Water hyacinth

C. Eurasian water milfoil

D. All of the above

27. Zebra mussels negatively affect freshwater ecosystems primarily by:

A. Increasing dissolved oxygen

B. Filtering out plankton and altering food webs

C. Raising water temperature

D. Increasing turbidity

28. Which organism is classified as moderately sensitive to pollution?

A. Damselfly

B. Tubifex worm

C. Leech

D. Blood midge

29. Short Answer: State one feeding habit of caddisfly larvae.

30. A stream sample dominated by midges and worms suggests:

A. Excellent water quality

B. Moderate water quality

C. Poor water quality

D. Recently restored habitat

Part III: Water Monitoring and Analysis (30 points)

31. A decrease in water temperature generally causes dissolved oxygen levels to:

A. Decrease

B. Increase

C. Stay the same

D. Fluctuate unpredictably

32. High biochemical oxygen demand (BOD) indicates:

A. Low microbial activity

B. High levels of organic pollution

C. High salinity

D. Low nutrient levels

33. Which parameter is most directly related to algal blooms?

A. Turbidity

B. Nitrates and phosphates

C. pH

D. Temperature

34. Short Answer: Explain how fecal coliform bacteria are used in water quality assessment.

35. Which unit is most commonly used to report turbidity?

A. ppm

B. mg/L

C. NTU

D. PSU

36. A river downstream of a wastewater discharge shows low DO and high nitrates. This most likely results from:

A. Thermal pollution only

B. Organic matter decomposition

C. Salinity intrusion

D. Reduced turbidity

37. Short Answer: Describe one relationship between turbidity and aquatic plant growth.

38. Which parameter is most affected by road salt runoff?

A. pH

B. Salinity

C. Dissolved oxygen

D. BOD

39. Total solids in water include:

A. Only dissolved materials

B. Only suspended materials

C. Both dissolved and suspended materials

D. Only organic materials

40. Short Answer: Why is long-term water monitoring more informative than a single measurement?

Part IV: Salinometer Testing (10 points)

Salinometer Construction (5 points) [ ] Device present and functional

Salinity Measurement (5 points) Estimate the salinity (%) of the provided solution to the nearest 0.1%

Measured Salinity: _____ %

BullSO, University of South Florida - Repaired practice edition

Division C | Water Quality | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (2 points) Authored written substitute for unavailable source material: Define “Point source” in the context of Water Quality.

Answer: _____
_____
_____